Task-1

```
varshini@Varshini:~$ pwd
/home/varshini
varshini@Varshini:~$ mkdir linux_task4
varshini@Varshini:~$ cd linux_task4
varshini@Varshini:~/linux_task4$ touch public.txt private.txt script.sh
varshini@Varshini:~/linux_task4$ echo "This is a public file." > public.tx
t
echo "This is a private file." > private.txt
echo '#!/bin/bash
echo "Hello from script"' > script.sh
varshini@Varshini:~/linux_task4$ ls -l
total 12
-rw-r--r-- 1 varshini varshini 24 Aug 12 08:31 private.txt
-rw-r--r-- 1 varshini varshini 23 Aug 12 08:31 public.txt
-rw-r--r-- 1 varshini varshini 37 Aug 12 08:31 script.sh
varshini@Varshini:~/linux_task4$ chmod 600 private.txt
varshini@Varshini:~/linux_task4$ chmod 644 public.txt
varshini@Varshini:~/linux_task4$ chmod +x script.sh
varshini@Varshini:~/linux_task4$ ls -l
total 12
-rw------- 1 varshini varshini 24 Aug 12 08:31 private.txt
-rw-r--r-- 1 varshini varshini 23 Aug 12 08:31 public.txt
-rwxr-xr-x 1 varshini varshini 37 Aug 12 08:31 script.sh
varshini@Varshini:~/linux_task4$ chmod -x script.sh
varshini@Varshini:~/linux_task4$ ls -l
total 12
-rw------- 1 varshini varshini 24 Aug 12 08:31 private.txt
-rw-r--r-- 1 varshini varshini 23 Aug 12 08:31 public.txt
-rw-r--r-- 1 varshini varshini 37 Aug 12 08:31 script.sh
varshini@Varshini:~/linux_task4$
```

Task-2:

```
varshini@Varshini:~/linux_task4$ mkdir permission_lab
cd permission_lab
varshini@Varshini:~/linux_task4/permission_lab$ touch test.txt
varshini@Varshini:~/linux_task4/permission_lab$ echo "This is a permission
 test file." > test.txt
varshini@Varshini:~/linux_task4/permission_lab$ chmod u-r test.txt
varshini@Varshini:~/linux_task4/permission_lab$ cat test.txt
cat: test.txt: Permission denied
varshini@Varshini:~/linux_task4/permission_lab$ chmod u+r test.txt
varshini@Varshini:~/linux_task4/permission_lab$ cat test.txt
This is a permission test file.
varshini@Varshini:~/linux_task4/permission_lab$ chmod u-w test.txt
varshini@Varshini:~/linux_task4/permission_lab$ echo "New content" >> test
.txt
-bash: test.txt: Permission denied
varshini@Varshini:~/linux_task4/permission_lab$ chmod u+w test.txt
varshini@Varshini:~/linux_task4/permission_lab$ echo "New content" >> test
.txt
varshini@Varshini:~/linux_task4/permission_lab$ chmod u-x test.txt
varshini@Varshini:~/linux_task4/permission_lab$ ./test.txt
-bash: ./test.txt: Permission denied
varshini@Varshini:~/linux_task4/permission_lab$
```

Task-3:

```
varshini@Varshini:~/linux_task4/permission_lab$ mkdir linux_task6
cd linux_task6
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ cat > important.txt << 'EOF'
Line 1: Linux is an operating system.
Line 2: Files can be copied using cp.
Line 3: Files can be moved using mv.
Line 4: Directories can be created using mkdir.
Line 5: Files can be removed using rm.
Line 6: Permissions can be changed using chmod.
Line 7: ls displays files and directories.
Line 8: cat displays file contents.
Line 9: head displays the beginning of a file.
Line 10: tail displays the end of a file.
EOF
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ mkdir backup
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ cp important.txt backup/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ mv backup/important.txt backup/important_backup.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ rm important.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ ls
backup
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ backup
Command 'backup' not found, but can be installed with:
sudo apt install openafs-client
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ cp backup/important_backup.txt .
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ mv important_backup.txt important_restored.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ cat important_restored.txt
Line 1: Linux is an operating system.
Line 2: Files can be copied using cp.
Line 3: Files can be moved using mv.
Line 4: Directories can be created using mkdir.
Line 5: Files can be removed using rm.
Line 6: Permissions can be changed using chmod.
Line 7: ls displays files and directories.
Line 8: cat displays file contents.
Line 9: head displays the beginning of a file.
Line 10: tail displays the end of a file.
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$
```

Task-4:

```
varshini@Varshini:~/linux_task4/permission_lab/linux_task6$ mkdir -p linux_project/source linux_project/backup linux_project/docs
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ touch source/main.c source/helper.c source/README.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ cat > source/README.txt << 'EOF'
Line 1: Linux Project
Line 2: Operating Systems
Line 3: File Management
Line 4: Linux Commands
Line 5: File Permissions
Line 6: Directory Management
Line 7: Copy Commands
Line 8: Move Commands
Line 9: Rename Commands
Line 10: Delete Commands
Line 11: Head Command
Line 12: Tail Command
Line 13: LS Command
Line 14: CP Command
Line 15: MV Command
Line 16: RM Command
Line 17: MKDIR Command
Line 18: CHMOD Command
Line 19: Backup Management
Line 20: End of README
EOF
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ head -n 5 source/README.txt
Line 1: Linux Project
Line 2: Operating Systems
Line 3: File Management
Line 4: Linux Commands
Line 5: File Permissions
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ tail -n 5 source/README.txt
Line 16: RM Command
Line 17: MKDIR Command
Line 18: CHMOD Command
Line 19: Backup Management
Line 20: End of README
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ cp source/main.c backup/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ cp source/README.txt docs/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ mv source/helper.c source/functions.c
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ chmod 644 source/README.txt
```

```
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ rm backup/main.c
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ rmdir backup
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ ls -R
.:
docs  source

./docs:
README.txt

./source:
README.txt  functions.c  main.c
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ tree
.
├── docs
│   └── README.txt
└── source
    ├── README.txt
    ├── functions.c
    └── main.c

3 directories, 4 files
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$
```

Task-5:

```
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project$ mkdir -p student_portal/src student_portal/docs student_portal/backup student_portal/logs student_portal/config
cd student_portal
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ touch main.c student.c README.txt config.txt system.log
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ mv main.c student.c src/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ mv README.txt docs/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ mv config.txt config/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ mv system.log logs/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ cat > logs/system.log << 'EOF'
Log 1: System started
Log 2: Student portal initialized
Log 3: Database connected
Log 4: User login successful
Log 5: Student record accessed
Log 6: Student details updated
Log 7: Course information loaded
Log 8: Attendance checked
Log 9: Marks retrieved
Log 10: Notification sent
Log 11: User logout successful
Log 12: Backup process started
Log 13: Backup process completed
Log 14: System check completed
Log 15: System running normally
EOF
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ head -n 5 logs/system.log
Log 1: System started
Log 2: Student portal initialized
Log 3: Database connected
Log 4: User login successful
Log 5: Student record accessed
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ tail -n 5 logs/system.log
Log 11: User logout successful
Log 12: Backup process started
Log 13: Backup process completed
Log 14: System check completed
Log 15: System running normally
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ cp docs/README.txt backup/
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ mv backup/README.txt backup/README_backup.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ chmod 600 config/config.txt
```

```
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ chmod 644 docs/README.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ rm backup/README_backup.txt
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ rmdir backup
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$ ls -R
.:
config  docs  logs  src

./config:
config.txt

./docs:
README.txt

./logs:
system.log

./src:
varshini@Varshini:~/linux_task4/permission_lab/linux_task6/linux_project/student_portal$
```